

AWS Cloud Technical Essentials: Configure a Web Application to use an Amazon S3 Bucket and Amazon DynamoDB Table

SPL-CA-100-CETENA-1 - Version 1.0.1

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at [AWS Training and Certification](#).

Lab overview

This lab guides you through configuring a web-based employee directory application by integrating Amazon S3 for photo storage and DynamoDB for employee data. You'll create and configure an S3 bucket to store employee images, set up appropriate bucket policies for access, and upload employee photos. Then you'll create a DynamoDB table to store employee information, and learn how to manage employee records through both the web interface and AWS Console. By the end of the lab, you'll have a fully functional employee directory that combines cloud storage and database services to manage employee information and images.

Objectives

After completing this lab, you will be able to:

- Create an Amazon Simple Storage Service (Amazon S3) Bucket
- Create an S3 bucket policy
- Modify an application to use an S3 Bucket
- Upload an Object to an S3 Bucket
- Create an Amazon DynamoDB table
- Test an application using an Application Web Interface
- Manage Existing DynamoDB Items Using the AWS Management Console
- Create Items in a DynamoDB Table Using the AWS Management Console

Technical knowledge prerequisites

This lab requires:

- Networked computer with Mac OS and Microsoft Windows, macOS, or Linux (Ubuntu, SUSE, or Red Hat)
- Administrator access (Microsoft Windows users)
- Internet browser, such as Chrome, Firefox, or Internet Explorer 9 or later

Note: Tablet devices cannot access the lab environment, although they can display student guides.

Icon key

Various icons are used throughout this lab to call attention to different types of instructions and notes. The following list explains the purpose for each icon:

- **Expected output:** A sample output that you can use to verify the output of a command or edited file.
- **Note:** A note, tip, or important guidance.
- **Additional information:** Where to find more information.
- **Consider:** A comment to consider how you might apply a concept in your own environment or to initiate a conversation about the topic at hand.
- **Copy/Paste:** A code block that displays the contents of a script or file you need to copy and paste that has been pre-created for you. When you need to copy only a certain part of a code block, there are numbered TODO comments in the code.

Start lab

1. To launch the lab, at the top of the page, choose **Start lab**.
2. **Caution:** You must wait for the provisioned AWS services to be ready before you can continue.
3. To open the lab, choose **Get started**.
 - You are automatically signed in to the AWS Management Console in a new web browser tab.
 - Warning:** Do not change the Region unless instructed.

Common sign-in errors

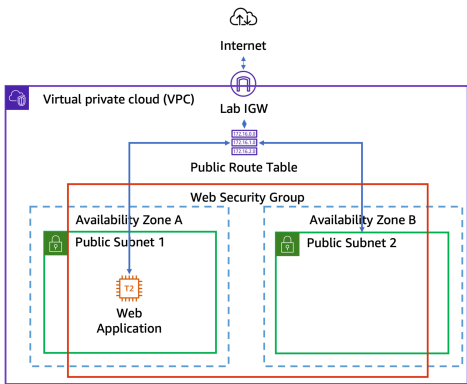
Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script Modern web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Task 1: Create an Amazon Simple Storage Service (Amazon S3) Bucket

To support this lab, we created some required resources for you. These resources include a VPC with two public subnets in two different Availability Zones, an Internet Gateway, a Route Table with a route to the Internet, and an EC2 instance hosting your employee directory application. See below for a resource overview.



Currently, the application runs in **Public Subnet 1**. The application does not have access to Amazon S3, but you provide that access later in this lab.

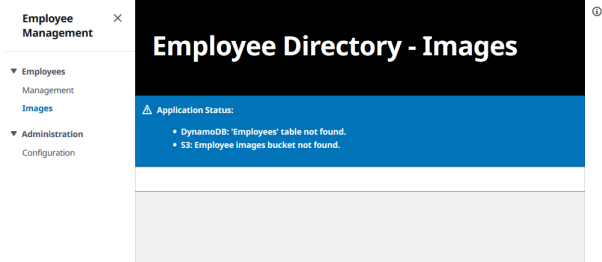
Applications must sign their API requests with AWS credentials to access other AWS resources. AWS Identity and Access Management (IAM) role enable applications to make secure API requests to an instance, without requiring you to manage the security credentials that the applications use.

Instead of creating and distributing AWS credentials for the application, you can delegate permission to make API requests using IAM roles. In this lab, the application uses the **EmployeeDirectoryAlphaRole** IAM role. The role hasn't been configured to allow Amazon S3 access. The application displays a warning message until Amazon S3 access is granted.

To access the employee directory application, do the following:

3. On the lab instructions page, in the left pane, copy **PublicApplicationURL**.
4. In a new browser tab, paste the URL you copied in the previous step. You should see the application, as shown.

Note: You might need to wait a few minutes before the web application becomes available.



A warning message states the following:

- S3: Employee Images bucket not found.
- DynamoDB: 'Employees' table not found.

First, you must address the S3 configuration. In the this, you create an S3 bucket and upload images to it. You also configure a policy to allow the application IAM role to access the S3 bucket, allowing the application to display the images.

In this task, you create an S3 bucket. Every object in Amazon S3 is stored in an S3 bucket. When you create your bucket, make sure that you create the bucket to your specific lab Region. You can find the lab region on the instructions page, in the left pane.

5. At the top of the AWS Management Console, in the search bar, search for and choose **S3**.
6. Choose **Create bucket**, and then configure the following:
 - Bucket name:** **enter your photo bucket -<INITIALS>-<NUMBER>**
 - Replace INITIALS with your initials.
 - Replace NUMBER with a random, four-digit number.
 - For Region,** make sure it matches the AWS Region value in the left pane of your instructions.

Example bucket name: **employee-photo-bucket-jef-1234**

Note: Each S3 bucket name must be globally unique.

When you select a particular Region, you can optimize latency, minimize costs, and address regulatory requirements, as needed. Objects stored in a Region never leave that Region, unless you explicitly transfer them to another Region.

The **Copy settings from an existing bucket** option creates a bucket that uses the same settings as another bucket. For this lab, you do not use this option.

7. In the **Block Public Access settings for this bucket** section, maintain the **Block all public access** section.
 - No changes are needed. The default setting is checked: **Block all public access.** This prevents all public access to data stored in the bucket.
8. Navigate to the bottom of the **Create bucket** page and choose **Create bucket**.

You should see a message similar to the following:

- Successfully created bucket "employee-photo-bucket-XXX-XXXX"**
 - Congratulations!** You successfully created an Amazon S3 Bucket for your employee directory web application images.
- In the next task, you create an Amazon S3 bucket policy.

Task 2: Create an S3 Bucket Policy

A bucket policy contains a set of permissions associated with an S3 bucket. The policy can control access to a entire bucket or to specific directories in a bucket.

9. Choose the **employee-photo-bucket-<INITIALS>-<NUMBER>** link for your bucket.
10. Choose the **Permissions** tab.
11. Navigate to the **Bucket policy** section, and then choose **Link**.

The S3 Management Console presents a sample **Bucket policy editor**. Bucket policies can be created manually or they can be created with the assistance of the **AWS Policy Generator**. In this lab, you create the policy manually.

12. Copy and paste the following policy into the **Bucket policy editor**:

```

{
  "Version": "2012-10-17",
  "Statement": [ {
    "Sid": "AllowS3ReadAccess",
    "Effect": "Allow",
    "Principal": {
      "AWS": "arn:aws:iam::<AWS-ACCOUNT-NUMBER>:role/EmployeeDirectoryAlpha"
    },
    "Action": "s3:*",
    "Resource": [
      "arn:aws:s3:::<BUCKET-NAME>",
      "arn:aws:s3:::<BUCKET-NAME>/*"
    ]
  } ]
}
```

13. Replace the two **<BUCKET-NAME>** placeholders with your bucket name, **employee-photo-bucket-<INITIALS>-<NUMBER>**.
14. In the left pane of the instructions, select and copy the value next to **AWSAccountID**.
15. Replace the **<ACCOUNT-NUMBER>** placeholder with the **AWSAccountID** value copied in the previous step.

Your **bucket policy** should look similar to the following example:

```

{
  "Version": "2012-10-17",
  "Statement": [ {
    "Sid": "AllowS3ReadAccess",
    "Effect": "Allow",
    "Principal": {
      "AWS": "arn:aws:iam::000000000000:role/EmployeeDirectoryAlpha"
    },
    "Action": "s3:*",
    "Resource": [
      "arn:aws:s3:::employee-photo-bucket-jef-1234",
      "arn:aws:s3:::employee-photo-bucket-jef-1234/*"
    ]
  } ]
}
```

16. Scroll to the bottom of the "Edit bucket policy" page and choose **Save changes**.

Note: If you receive an error regarding the first byte, ensure there is no empty space before the first (character on line 1. If there is, delete the empty space and try again.

If your policy is correct, you should receive a message similar to the following:

- Successfully edited bucket policy**

You applied a bucket policy to your S3 bucket. The bucket policy uses the **EmployeeDirectoryAlphaRole** IAM role to allow read access from your application to the S3 bucket. With this policy, all objects in your bucket are accessible to your application.

- Congratulations!** You successfully created a bucket policy for your Amazon S3 Bucket.

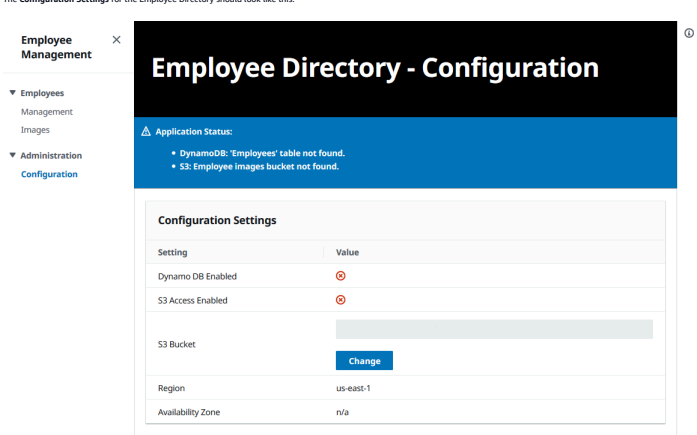
In the next task, you modify the employee directory applications to use the Amazon S3 bucket.

Task 3: Modify the Application to Use the S3 Bucket

In this task, you configure your application to use the bucket as a source for employee images.

17. Return to the browser tab with the Employee Directory application.
18. In the **Administration** section, choose the **Configuration**.

The **Configuration Settings** for the Employee Directory should look like this:



The **S3 Access Enabled** value and blank **S3 Bucket** value indicate an S3 bucket has not been associated with the Employee Directory application.

Use the **Configuration Settings** section to associate your S3 bucket (**employee-photo-bucket-<INITIALS>-<NUMBER>**) with the application.

19. Choose **Change** in the **S3 Bucket** field.
20. Enter your bucket name (i.e. **employee-photo-bucket-<INITIALS>-<NUMBER>**) in the **S3 Bucket** field.
21. Choose **Save**.

Your **Configuration Settings** page should now look like the following:

Setting	Value
DynamoDB Enabled	
S3 Access Enabled	
S3 Bucket	employee-photo-bucket-jef-1234
Region	us-east-1
Availability Zone	us-east-1a
EC2 Instance Id	i-016b1563d436b073

The **S3 Access Enabled** value and blank **S3 Bucket** value indicate an S3 bucket has been successfully associated with the employee directory web application.

- Congratulations!** You successfully configured your employee directory web application to use an Amazon S3 Bucket to host your employee images.

In the next task, you upload the employee images to the Amazon S3 bucket.

Task 4: Upload an Object to an S3 Bucket

